SUPERSEDING
MIL-I-27674A(USAF)
8 February 1963

MILITARY SPECIFICATION

INDICATORS, TEMPERATURE, ELECTRICAL RESISTANCE MULTIFUNCTION, -50° TO +200°C

1. SCOPE

1.1 This specification covers the requirements for multifunction electrical resistance temperature indicators.

1.2 Classification. Indicators shall be of the following types, as specified (see 6.2):

EHU-21A - Antireflective coating on coverglass per MIL-C-675 with white integral lighting.

EHU-21A/A - High efficiency antireflective coating on coverglass with white integral lighting.

EHU-21A/A-1 - High efficiency antireflective coating on coverglass with red integral lighting.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-C-675	Coating Of Glass Optical Elements (Anti-Reflection)
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-I-7070	Indicator, Temperature, Electrical Resistance, Hermetically Sealed, General Specification For
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White General Specification For

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-785	Reliability Program For Systems And Equipment Development And Production
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FEDERAL AVIATION AGENCY

Technical Standard Order

TSO-C43	Temperature Indicators

(Copies of the FAA document may be obtained from the Federal Aviation Agency, Washington DC 20553.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago IL 60606.)

3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 General specification. The requirements of MIL-I-7070 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

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3.3 Construction.

3.3.1 Case. The indicator case shall be in accordance with MS33639 for the 2-inch diameter size. The case length shall be 4 ±1/2 inches.

3.3.2 Dial. The dial shall conform to Figure 1 for EHU-21/A and EHU-21A/A; for EHU-21A/A-1 "CSD" (constant speed drive) shall be replaced by "WBT" (weapons bay temperature). The diameter of the dial as measured across the outer ends of the graduations shall be not less than 1-3/4 inches. The 150° Centigrade (C) indication shall be positioned on the horizontal centerline at the left side of the dial. The dial blank shall consist of a flat disc of plastic, metal, or other material which will permit compliance with the lighting and other requirements specified herein.

3.3.2.1 Lighting. The dial shall be integrally lighted in accordance with MIL-L-27160 or MIL-L-25467, as applicable.

3.3.2.2 Scale arc. The indicator shall provide a scale arc angle of 260 degrees ±1 degree. The mechanism shall not be geared to obtain the specified scale arc. Adjacent scale graduations shall be spaced within 1 angular degree of precise linear spacing.

3.3.2.3 Coverglass coating. The reflecting glass surface of the EHU-21/A indicator shall be finished with a coating which conforms to MIL-C-675. The EHU-21A/A and EHU-21A/A-1 indicator shall be finished with a high efficiency coverglass coating conforming to the following. The coatings shall withstand the environmental conditions specified herein.

Reflectance Tolerances

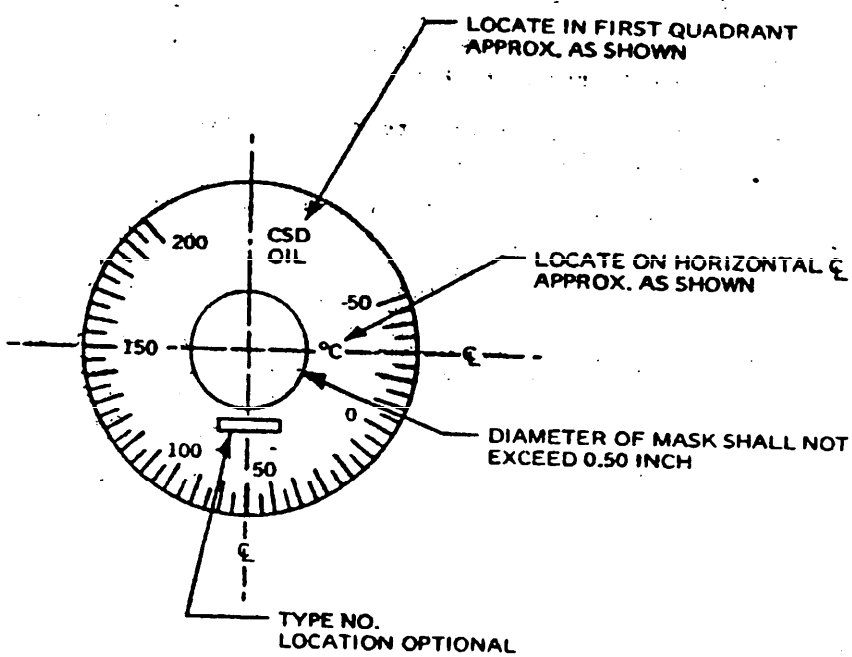
Angle of Incidence	Wavelength Millimicrons	Percent Reflectance
0 degree	450 - 675	0.6 maximum
0 degree	425 - 700	0.5 average
30 degrees	450 - 625	1.0 maximum
30 degrees	425 - 700	0.5 average

3.3.3 Pointer. The pointer shall be lighted in accordance with MIL-L-27160. The pointer width 0.50 ±0.063 inch from the tip shall be 0.078 ±0.016 inch and shall taper to a width of 0.020 ±0.005 inch at the tip. The length of the pointer shall be such that the tip of the pointer will extend 0.019 ±0.016 inch into the shortest scale graduation.

3.3.3.1 Pointer overswing. The pointer overswing shall be not less than 3 angular degrees beyond each end of the scale.

3.4 Dial to coverglass location. The distance between the inner surface of the coverglass and the dial surface shall not exceed 0.198 inch.

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NOTE: "CSD OIL" SHALL BE REPLACED BY "WBT" FOR EHU-21A/A-1

FIGURE 1. INDICATOR

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3.5 Electrical receptacle. An MS33678-14S-5P hermetically sealed electrical receptacle shall be provided. It shall be located on the vertical and horizontal centerlines at the rear of the case. The polarizing pin shall be located at top center of the receptacle. The receptacle shall be wired as follows:

Pin	Connection
A	28 volts direct current (DC)
B	Resistance bulb connection
C	Ground and remaining bulb connection
D	5 volts lighting input
E	5 volts lighting ground

3.6 Electronic components. Electronic components and the application thereof shall conform to MIL-E-5400.

3.7 Power consumption. The power consumption of the indicator, exclusive of the lighting circuit, shall not exceed 2 watts.

3.8 Markings.

3.8.1 Dimensions and colors. Dimensions and colors of the markings shall be as follows. The colors shall be in accordance with FED-STD-595.

Marking	Height or Length Inch ±0.005	Width of Line or Graduation Inch ±0.005	Color No.
Numerals	0.156	0.031	37875
Major graduations	0.156	0.031	37875
Minor graduations	0.094	0.020	37875
Lettering "°C"	0.125	0.020	37875
Oil CSD	0.094	0.020	37875
Background of dial and of mask	-----	-----	37038
Type No.	0.063	0.010	37038

3.9 Nameplate. A metal nameplate shall be permanently attached on the rear of the indicator case. Metal cals shall not be used.

3.10 Weight. The weight of the indicator shall not exceed 1.0 pound.

* 3.11 Reliability. The contractor shall establish a reliability program in accordance with MIL-STD-785.

* 3.11.1 The indicator shall have a specified mean time between failures (MTBF) of 1000 hours.

3.12 TSO conformance. The indicator shall conform to the requirements of Federal Aviation Agency TSO-C43 for type I indicators.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General specification. The inspection and testing of indicators shall be in accordance with MIL-I-7070, except as specified herein.

4.3 Test conditions.

4.3.1 Test voltage. Unless otherwise specified, the test voltage shall be 28.5 volts DC and the lighting circuit shall be energized with 4.5 ±0.5 volts during all tests.

4.3.2 Test tolerances. All test tolerances shall be in accordance with Table I. The tolerances not enclosed in parentheses are applicable only when the indicators are subjected to all the tests in the sequence specified, and the tolerances enclosed in parentheses are applicable only when the indicators are subjected to a specific test, independent of sequence.

4.4 Test methods.

4.4.1 Individual tests.

4.4.1.1 Scale error at room temperature. The test resistance specified in Table I shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors at the test points shall not exceed the tolerance specified in Table I.

4.4.1.2 Position error. The position error of the indicator shall not exceed 2.5°C.

4.4.1.3 Friction error. The friction error of the indicator shall not exceed 2°C.

4.4.1.4 Insulation leakage. The insulation resistance shall be a minimum of 20 megohms.

4.4.1.5 The indicator shall be subjected to the individual tests specified in MIL-L-27160.

4.4.1.6 Sealing. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium-leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere. Where 100 percent helium is utilized as the filling medium, the detected leak rate shall not exceed 1.0 micron cubic foot per hour.

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TABLE I

TEST TOLERANCES

Test Point	Test Resistance	Scale Error at Room Temperature	Reduced Voltage	Scale Error at Low Temperature	Scale Error at High Temperature	Scale Error after Vibration
°C	Ohms	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C
-50	74.24	4.0 (4.0)	3.0 (3.0)	6.0 (6.0)	6.0 (6.0)	6.0 (6.0)
-30	80.56	3.0 (3.0)	3.0 (3.0)	5.0 (5.0)	5.0 (5.0)	6.0 (5.0)
0	90.38	3.0 (3.0)	3.0 (3.0)	5.0 (5.0)	5.0 (5.0)	6.0 (5.0)
+30	100.91	3.0 (3.0)	3.0 (3.0)	5.0 (5.0)	5.0 (5.0)	6.0 (5.0)
80	120.36	2.0 (2.0)	2.0 (2.0)	4.0 (4.0)	4.0 (4.0)	5.0 (4.0)
120	137.78	2.0 (2.0)	2.0 (2.0)	4.0 (4.0)	4.0 (4.0)	5.0 (4.0)
150	151.91	2.0 (2.0)	2.0 (2.0)	4.0 (4.0)	4.0 (4.0)	5.0 (4.0)
200	177.95	4.0 (4.0)	3.0 (3.0)	6.0 (6.0)	6.0 (6.0)	6.0 (6.0)

Test Point	Test Resistance	Open and Short Circuit	Bulb Lead Resistance	EXTREME TEMPERATURE EXPOSURE		
				Scale Error at Low Temprature	Room Temperature After Low Temperature	Room Temperature After High Temperature
°C	Ohms	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C
-50	74.24	7.0 (5.0)	2.0 (2.0)	9.0 (6.0)	7.0 (4.0)	7.0 (4.0)
-30	80.56	6.0 (4.0)	2.0 (2.0)	8.0 (5.0)	6.0 (3.0)	6.0 (3.0)
0	90.38	6.0 (4.0)	2.0 (2.0)	8.0 (5.0)	6.0 (3.0)	6.0 (3.0)
+30	100.91	6.0 (4.0)	2.0 (2.0)	8.0 (5.0)	6.0 (3.0)	6.0 (3.0)
80	120.36	5.0 (3.0)	2.0 (2.0)	7.0 (4.0)	5.0 (2.0)	5.0 (2.0)
120	137.78	5.0 (3.0)	2.0 (2.0)	7.0 (4.0)	5.0 (2.0)	5.0 (2.0)
150	151.91	5.0 (3.0)	2.0 (2.0)	7.0 (4.0)	5.0 (2.0)	5.0 (2.0)
200	177.95	7.0 (5.0)	2.0 (2.0)	9.0 (6.0)	7.0 (4.0)	7.0 (4.0

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4.4.2 Sampling plan A tests. Samples shall be selected in accordance with MIL-STD-105, AQL of 2.5, inspection level II.

4.4.2.1 Reduced voltage operation. The indicator shall be subjected to a scale error at room temperature test with an applied voltage of 22.5 volts at the test points in Table I. The indications at the reduced voltage shall not differ from the indications with 28.5 volts applied by more than the tolerances specified in Table I.

4.4.2.2 Scale error at low temperature. The scale errors at low temperature shall not exceed the tolerances specified in Table I.

4.4.2.3 Scale error at high temperature. The scale errors at high temperature shall not exceed the tolerances specified in Table I.

4.4.2.4 Magnetic effect The magnetic compass shall not deflect more than 5 degrees.

4.4.2.5 Resistance bulb current. A total resistance of 100 ohms shall be applied across the resistance bulb terminals of the indicator, the test voltage applied, and the resistance bulb circuit current measured. The current shall not exceed 22 milliamperes.

4.4.2.6 Open and short circuit. After open and short circuit, the scale errors shall not exceed the tolerances specified in Table I.

4.4.2.7 Vibration. The indicator shall be subjected to vibration in accordance with procedure XII, curve B of MIL-E-5272. Pointer oscillation or pointer variation shall not exceed 2.5°C. The test shall be conducted with power applied and with input resistance to provide an indication of 150°C. The scale errors at room temperature after vibration shall not exceed the tolerances specified in Table I. The resonance and cycling portions of this test shall be conducted through the specified range at vibratory accelerations not exceeding those shown for curve B. When pointer oscillation and pointer variation are noted, the vibration frequency shall be 5 to 500 cycles per second at double amplitudes that will produce a vibration level of 1.5g maximum.

4.4.2.8 The sampling plan A tests specified in MIL-L-27160 or MIL-L-25467, as applicable, shall be conducted.

4.4.3 Sampling plan B tests. Samples shall be selected in accordance with MIL-STD-105, AQL of 6.5, special inspection level S-1.

4.4.3.1 Response time. The response time shall not exceed 3 seconds.

4.4.3.2 Damping factor. The damping factor shall be not less than 1.6.

4.4.3.3 Power consumption. The power consumption shall not exceed 2 watts.

4.4.3.4 Bulb lead resistance. The indicator shall be connected as shown on Figure 2 and subjected to a scale error at room temperature test. The indications with bulb lead resistance shall not differ from the indications without bulb lead resistance by more than the tolerances specified in Table I.

4.4.3.5 Extreme low temperature exposure. After low temperature exposure, the scale errors at low temperature and the scale errors at room temperature shall not exceed the tolerances specified in Table I.

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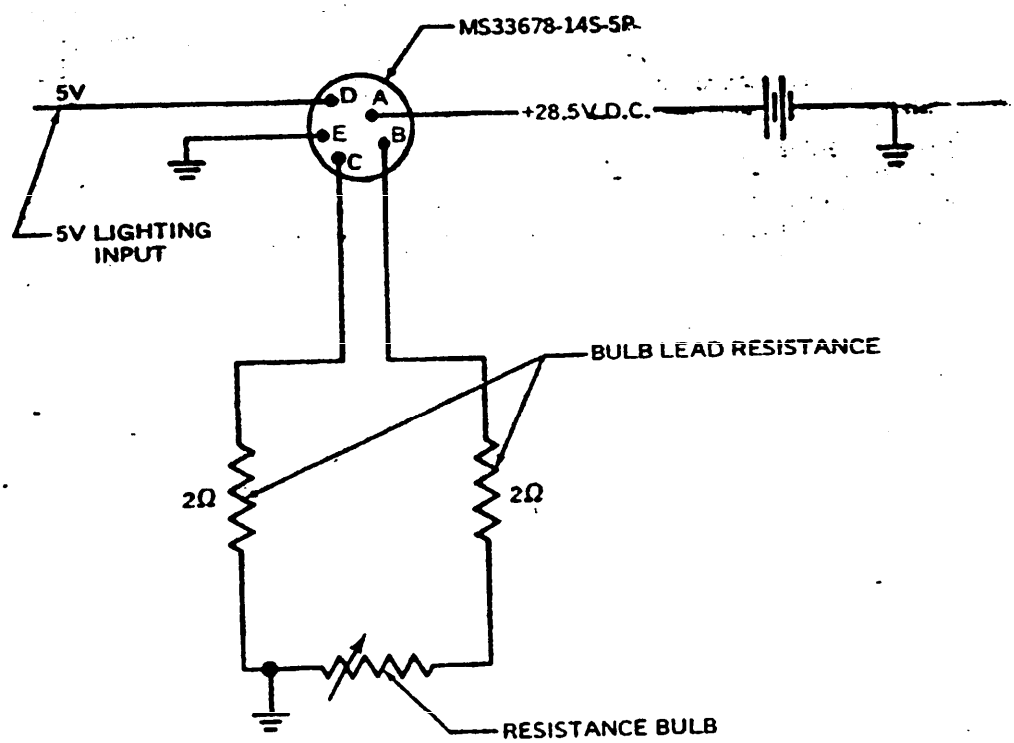


FIGURE 2. WIRING DIAGRAM

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4.4.3.6 High temperature exposure. After high temperature exposure, the scale errors at room temperature shall not exceed the tolerances specified in Table I.

4.4.3.7 Salt spray. The indicator shall be subjected to and shall meet the sealing test following salt spray exposure. The insulation resistance shall be a minimum of 20 megohms.

4.4.3.8 The sampling plan B tests specified in MIL-L-27160 or MIL-L-25467, as applicable, shall be conducted.

4.4.3.9 Thermal shock. No evidence of moisture penetration or damage to the inclosure shall result from this test. Following this test, the indicator shall be resubjected to and meet the test specified in 4.4.1.6. Fogging of the coverglass during any portion of this test shall be cause for rejection.

* 4.4.3.10 Reliability demonstration. The qualification test samples shall first be subjected to the scale error test and then to cycles consisting of 4 hours of operation with power and resistance applied and 1 hour with power removed. The cycles shall be conducted at the following ambient temperatures, ±2°C, in the sequence specified: 4 hours at 0°C, 1 hour off; 4 hours at room temperature, 1 hour off; 4 hours at 50°C, 1 hour off. During the operational portion of the cycles, the resistance shall be so varied as to produce an indication cycle corresponding to a change from -50° to +150°C and return to -50°C. This shall constitute 1 indication cycle which shall be repeated at a rate of 6 ±1 cycle per minute. The continuity of this test may be interrupted for weekend shutdown. A minimum of three and a maximum of five sample indicators shall be tested by the contractor for reliability. Duration of the test and accept-reject criteria shall be as specified in MIL-STD-781, Test Plan III. At the conclusion of this test, the indicators shall be subjected to the scale error at room temperature and the friction error tests. The scale errors at the respective test points shall not exceed 150 percent of the scale errors prior to this test, and the friction error shall not exceed 3°C. Reliability of the equipment will be verified when the indicators have passed all tests specified herein.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be levels A or C, as specified (see 6.2).

* 5.1.1 Level A. Each indicator shall be preserved and packaged in accordance with MIL-P-116, Method III and placed in a snug fitting container conforming to Grade W6C, PPP-B-636.

* 5.1.2 Level C. Each indicator shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. The supplier may use his commercial practice providing it meets this requirement.

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* 5.2 Packing. Packing shall be levels A, B, or C, as specified (see 6.2).

* 5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in weather resistant shipping containers conforming to PPP-B-636. The grade of each container determined by the gross weight and size, shall comply to the applicable special requirements table of the specification. Waterproofing, closure and strapping shall be in accordance with the container specification. Containers shall be of uniform shape and size and of minimum cube and tare.

* 5.2.2 Level B. Same as level A, except that waterproofing is not required.

* 5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior type shipping containers in a manner that will insure safe transportation at lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules, or regulations of other carriers applicable to the mode of transportation.

* 5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use in indicating temperatures, such as constant speed drive oil temperatures, ranging from -50° to +200°C. The indicators may be used on instrument panels by application of an MS28042-1 instrument mounting clamp.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Type of indicator required (see 1.2).

c. Level of packaging and packing desired.

d. Whether sampling plan B tests are to be conducted.

6.3 Definitions. Definitions will be as specified in MIL-I-7070.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Materiel Area, Service Engineering Division, ATTN: MMEC, Kelly AFB, Texas 78241, and information pertaining to qualification of products may be obtained from that activity.

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* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 82

Review Activity:
Air Force - 11

Preparing Activity:
Air Force - 82

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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